

RV UNIVERSITY, BENGALURU-59

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

A Project Synopsis On

PREDICTIVE WASTE MANAGEMENT SYSTEM

USING IoT

For the Course Internet of Things (CS2404)

Submitted By

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ABSTRACT

This project is a smart dustbin that will monitor the waste level in it. It also weighs the trash. This information is collected by small devices placed in the bin. The information is displayed on a small screen. The green, yellow and red lights indicate the bin status. When the bin is almost full a buzzer gives sound. The system can also use previous readings to get an idea of when the bin may be full. The main objective is to collect the waste at the right time and not allow bins to overflow. The project is straightforward and inexpensive.

1. Introduction

Waste is typically collected at scheduled times. Sometimes a bin gets full before it is picked up. Sometimes when it is collected, the bin is almost empty. This can be inefficient and time consuming. Our smart dustbin regularly checks the dustbin . It shows the fullness of the bin and the amount of waste in it. It can also use past information to estimate when the bin might be full. This helps the workers in collecting the waste before the bin overflows.

2. Goals

- Check how much waste is in the bin.
- Check the approximate weight of the garbage.
- display the info on a small screen
- Provide a light and sound warning when the bin is filling up.
- Using the information you found earlier, estimate when the bin might be full.
- Economically build the project.

3. The Operation of the Project

First the devices check the waste bin. One device checks the empty space inside the bin, another checks the weight and the other devices check temperature, humidity and air condition. This information is received by the main board the ESP32. Results are shown on screen. The lights show the status of the bin. The buzzer warns you when the bin is almost full. The system can also check out older readings and tell you roughly when the bin will be full.

4. Parts Utilized

- Part What it does Main board ESP32. All the parts send information to it.
- HC-SR04 Checks for remaining empty space in the bin.
- Load Cell + HX711 Measures the approximate weight of the waste.
- DHT11 Check temperature and humidity.
- MQ-135 Checks the air condition in the container.
- OLED Screen Displays readings and bin condition.
- Red, Yellow, Green Lights tell if the bin is okay, filling up or full.
- Buzzer Beeps when the bin needs attention.
- Breadboard + Wires To connect the parts together.
- Dustbin / Box A small bin used to demonstrate the working of the project.

5. Approximate Cost

No.	Part	Qty.	Total (Rs.)
1	ESP32	1	413
2	HC-SR04	1	83
3	Load Cell + HX711	1	295
4	DHT11	1	94
5	MQ-135	1	118
6	OLED Screen	1	142
7	Lights	3	18
8	Buzzer	1	24
9	Breadboard + wires	1 set	177
10	Dustbin / box	1	118
			1,300

6. Simple Working Flow

Check the bin → **ESP32 receives the information** → **Screen shows the result** → **Lights and buzzer**

show a warning → **Earlier information helps estimate when the bin may be full.**

7. Anticipated Results

With the completion of the project, we expect the project to be able to:

- Indicate the fullness of the bin.
- Indicate the approximate weight of the garbage.
- Indicate the air quality, temperature and humidity in the location of the bin.
- Indicate the condition of the bins on the display.
- Indicate the fullness of the bin with the help of red, green and yellow lights on the given dustbin.
- Indicate the time at which the bin will be full.
- Provide with the idea of how smart garbage bins can assist in waste collection

8. Benefits

- * Aids in preventing overflowing bins.
- * Can eliminate unnecessary collection trips.
- * There is no complex functionality.
- * Simple and cheap parts are used.
- * Can be utilized as a small campus project.

9.Literature Survey

Several researchers have proposed IoT-based solutions for improving waste management. BIN-CT (2019) focused on predicting the fill level of waste containers using data and machine-learning techniques mainly to improve waste collection and route planning. However the work is primarily focused on prediction and city-level collection optimization.

An Based Smart Waste Management System Using LoRa and TensorFlow (2020) introduced sensors, communication technology and deep learning for monitoring and classifying waste but the system requires several components and has a relatively complex architecture.

Smart Waste Management Scheme using IoT for Metropolitan Cities (2023) proposed IoT-enabled garbage bins for monitoring waste levels and improving collection efficiency. Mainly addressed large-scale metropolitan waste management rather than a simple low-cost prototype

A Smart Bin with Real-Time Monitoring and Garbage Level Tracking using IoT (2024) used weight sensors to monitor the amount of waste in real time. While it provides monitoring its main focus is on the current status of the bin rather than predicting when it will become full.

Recently Predictive Analysis-Based Sustainable Waste Management using IoT Edge Computing and Blockchain (2025) combined IoT, edge computing and machine-learning techniques for waste prediction and collection scheduling but its architecture is comparatively complex for a small-scale implementation.

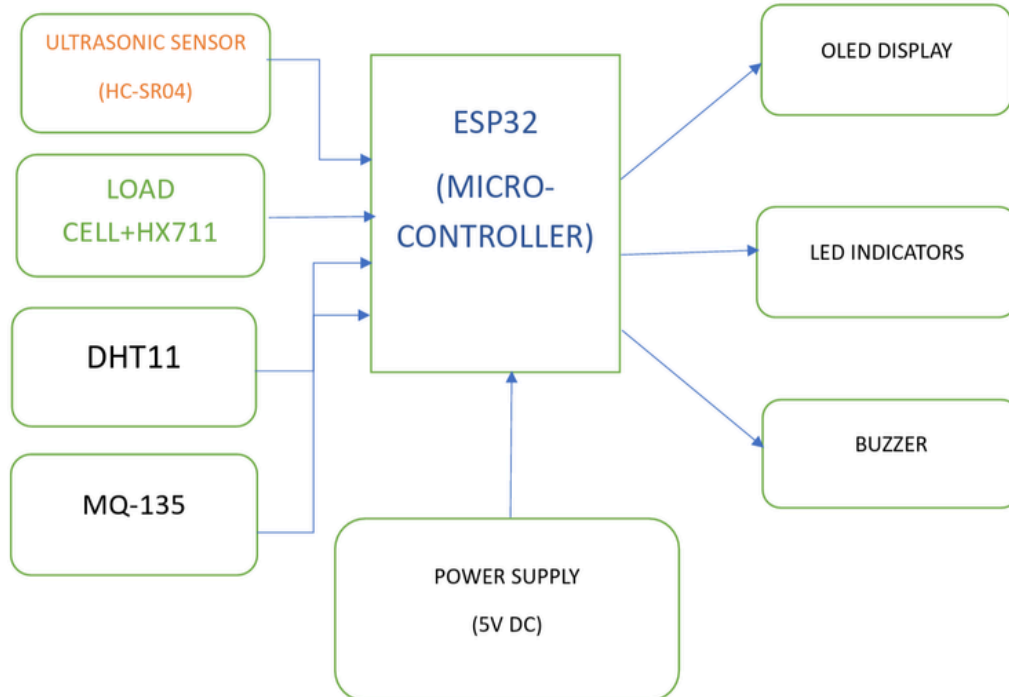
Based on these studies there is an opportunity to develop an low-cost predictive smart-bin system that combines multiple sensor parameters with local monitoring and fullness prediction. Our proposed system addresses this by using an ESP32 with fill-level, weight, temperature, humidity and air-quality sensors along with an OLED, LEDs and buzzer for status indication. This one also aims to keep the design affordable and scalable for communities or pilot projects. It uses processing to reduce dependency on cloud services and allows faster response times through real-time alerts.

The system not tracks the current fill level but also predicts when the bin will reach capacity based on usage patterns. This helps optimize collection schedules without requiring human oversight. It supports detection of issues like high temperature or poor air quality inside the bin, which could indicate fire risks or biological decomposition problems. By integrating sensors and using lightweight machine-learning models it delivers accurate predictions while staying cost-effective. The goal is to create a solution that works well in both urban and rural areas where waste management resources are limited. This system can be easily installed in spaces, residential buildings or small commercial zones. Its modular design allows upgrades, such, as adding solar power or connectivity options. Overall this approach brings together simplicity, affordability and intelligence to make waste management smarter and more sustainable.

No.	Research Paper	Year	Already Done	Research Gap	Our Project
1	BIN-CT: Urban Waste Collection Based on Predicting Container Fill Level	2019	Fill-level prediction and collection optimization	Mainly prediction/route planning	Adds multi-sensor monitoring + local alert + prediction
2	IoT-Based Smart Waste Management using LoRa and TensorFlow	2020	IoT monitoring, waste classification and prediction	Complex and costly architecture	Low-cost ESP32-based implementation
3	Smart Waste Management Scheme using IoT for Metropolitan Cities	2023	IoT-enabled waste-bin monitoring	Focused mainly on large-scale city systems	Suitable for small/campus-scale implementation
4	Smart Bin with Real-Time Monitoring and Garbage Level Tracking	2024	Real-time fill-level and weight monitoring	Mainly monitors present condition	Adds prediction of future fullness
5	Predictive Analysis-Based Sustainable Waste Management using IoT Edge Computing and Blockchain	2025	Prediction, IoT, edge computing and collection scheduling	Complex architecture with blockchain/federated systems	Simpler predictive edge-IoT approach

INPUT SENSORS

OUTPUT SENSORS



9. Conclusion

The Predictive Waste Management System is an uncomplicated smart dustbin project. It monitors the volume and weight of waste and sends an alert when the bin is nearly overflowing. It also analyzes the previous data meaning it helps predict when the bin is going to be full. The project is simple to present and inexpensive at the same time.

Reference:

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